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Operations Research & Optimization

2026 7 30

- 1 (Linear Programming)
- 2 (Nonlinear Programming)
- 3 (Convex Optimization)
- 4 ML

(LP)

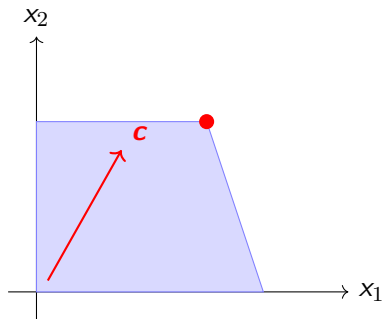
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- /
- LP

(Standard Form)

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$

$$\mathbf{A} \in \mathbb{R}^{m \times n} \quad \mathbf{b} \in \mathbb{R}^m \quad \mathbf{c} \in \mathbb{R}^n$$

- **(Polyhedron)** $\{x \mid Ax = b, x \geq 0\}$
- (Convex Polytope)
- **(extreme point / basic feasible solution)**
- (Simplex Method)



LP

- $\max \mathbf{c}^\top \mathbf{x} \iff \min -\mathbf{c}^\top \mathbf{x}$
- $\mathbf{Ax} \leq \mathbf{b} \iff \mathbf{Ax} + \mathbf{s} = \mathbf{b}, \mathbf{s} \geq \mathbf{0}$
- $\mathbf{Ax} \geq \mathbf{b} \iff \mathbf{Ax} - \mathbf{s} = \mathbf{b}, \mathbf{s} \geq \mathbf{0}$
- $x_i \quad \iff \quad x_i = x_i^+ - x_i^-, x_i^+, x_i^- \geq 0$

LP

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Simplex

(Basic Feasible Solution, BFS) $A \quad m \quad B \quad 0 \quad Bx_B = b$

- 1 BFS
- 2 (reduced cost) $\bar{c}_N^\top = c_N^\top - c_B^\top B^{-1}N$
- 3 $\bar{c}_N \geq 0$
- 4 Dantzig (ratio test)
- 5

(Simplex Tableau)

LP

	$\mathbf{x}_B$	$\mathbf{x}_N$	RHS
$\mathbf{x}_B$	$\mathbf{I}$	$\mathbf{B}^{-1}\mathbf{N}$	$\mathbf{B}^{-1}\mathbf{b}$
z	$\mathbf{0}$	$\bar{\mathbf{c}}_N^\top$	$\mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{b}$

- $\bar{\mathbf{c}}_N \geq \mathbf{0}$
- $\bar{c}_j < 0 \leq \mathbf{0}$
- Min ratio test

(Two-Phase Method)

BFS

■ **Phase I** $\mathbf{a}$

$$\min \sum_i a_i \quad \text{s.t.} \quad \mathbf{Ax} + \mathbf{a} = \mathbf{b}, \quad \mathbf{x}, \mathbf{a} \geq \mathbf{0}$$

> 0

■ **Phase II** Phase I BFS

■ M $M \sum a_i$

(Primal)

$$\min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$$

(Dual)

$$\max_{\mathbf{y}} \mathbf{b}^\top \mathbf{y} \quad \text{s.t.} \quad \mathbf{A}^\top \mathbf{y} \leq \mathbf{c}$$

Primal ()	Dual ()
$x_i \geq 0$	$\mathbf{a}_i^\top \mathbf{y} \leq c_i$
x_i	$\mathbf{a}_i^\top \mathbf{y} = c_i$
$\mathbf{a}_i^\top \mathbf{x} \geq b_i$	$y_i \geq 0$
$\mathbf{a}_i^\top \mathbf{x} = b_i$	y_i

(Weak Duality)

$\mathbf{x}$ Primal $\mathbf{y}$ Dual

$$\mathbf{c}^\top \mathbf{x} \geq \mathbf{b}^\top \mathbf{y}$$

(Strong Duality)

Primal Dual

$$\min \mathbf{c}^\top \mathbf{x} = \max \mathbf{b}^\top \mathbf{y}$$

(Complementary Slackness)

$$x_i^* \cdot (\mathbf{a}_i^\top \mathbf{y}^* - c_i) = 0 \quad i$$

$$y_i^* \cdot (\mathbf{a}_i^\top \mathbf{x}^* - b_i) = 0 \quad i$$

- y (Shadow Price)

- $y_i = \frac{\partial \text{Optimal Value}}{\partial b_i}$

- $y_i \geq 0$

- $y_i > 0$

(Sensitivity Analysis) b_i, c_j

- $B^{-1}b \geq 0$

- $\bar{c}_N \geq 0$

LP (Interior Point Method)

- KKT

$$\begin{aligned} \mathbf{Ax} &= \mathbf{b}, \quad \mathbf{x} > \mathbf{0} \\ \mathbf{A}^\top \mathbf{y} + \mathbf{s} &= \mathbf{c}, \quad \mathbf{s} > \mathbf{0} \\ \mathbf{XS}\mathbf{e} &= \mu \mathbf{e} \end{aligned}$$

$$\mathbf{X} = \text{diag}(\mathbf{x}), \quad \mathbf{S} = \text{diag}(\mathbf{s}) \quad \mu \rightarrow 0$$

Newton μ LP

-
- $m \quad n$
- Hungarian $n \quad n$
- $/ \rightarrow$ LP
-
- **(DEA)**

LP Gurobi, CPLEX, HiGHS, GLPK

NLP

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p\end{array}$$

f, g_i, h_j

- **NLP** $\min f(\mathbf{x})$
- **NLP**
- $\rightarrow$

$$\min f(\mathbf{x}), \mathbf{x} \in \mathbb{R}^n$$

$$\mathbf{x}^* \quad f \quad \nabla f(\mathbf{x}^*) = \mathbf{0}$$

$$\mathbf{x}^* \quad f \quad \nabla f(\mathbf{x}^*) = \mathbf{0} \quad \nabla^2 f(\mathbf{x}^*) \succeq \mathbf{0}$$

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \quad \nabla^2 f(\mathbf{x}^*) \succ \mathbf{0} \quad \mathbf{x}^*$$

(Gradient Descent)

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k)$$

- α_k (step size / learning rate)
- $\mathcal{O}(1/k)$
- **Wolfe**
- **Backtracking line search**

ill-conditioned

(Newton's Method)

Taylor

$$f(\mathbf{x}_k + \mathbf{p}) \approx f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^\top \mathbf{p} + \frac{1}{2} \mathbf{p}^\top \nabla^2 f(\mathbf{x}_k) \mathbf{p}$$

Newton $\mathbf{p}_k = -[\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$$



Hessian $\mathcal{O}(n^3)$ Hessian

(Quasi-Newton Methods)

Hessian Hessian

BFGS (Broyden–Fletcher–Goldfarb–Shanno)

$$s_k = \mathbf{x}_{k+1} - \mathbf{x}_k$$

$$y_k = \nabla f(\mathbf{x}_{k+1}) - \nabla f(\mathbf{x}_k)$$

$$\mathbf{H}_{k+1} = \left(\mathbf{I} - \rho_k s_k y_k^\top \right) \mathbf{H}_k \left(\mathbf{I} - \rho_k y_k s_k^\top \right) + \rho_k s_k s_k^\top$$

$$\rho_k = 1 / (y_k^\top s_k)$$



■ **L-BFGS** m (s_k, y_k) $\mathcal{O}(mn)$

Karush–Kuhn–Tucker (KKT)

NLP — KKT

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \mu_i \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \lambda_j \nabla h_j(\mathbf{x}^*) = \mathbf{0}$$

$$g_i(\mathbf{x}^*) \leq 0, \quad i = 1, \dots, m$$

$$h_j(\mathbf{x}^*) = 0, \quad j = 1, \dots, p$$

$$\mu_i \geq 0, \quad i = 1, \dots, m$$

$$\mu_i \cdot g_i(\mathbf{x}^*) = 0, \quad i = 1, \dots, m \quad (\quad)$$

- μ_i Lagrange
- λ_j Lagrange
- KKT

- **(Penalty Method)**

$$\min_{\mathbf{x}} f(\mathbf{x}) + \rho \left[\sum_i \max(0, g_i(\mathbf{x}))^2 + \sum_j h_j(\mathbf{x})^2 \right]$$

$\rho \rightarrow \infty$ ill-conditioning

- **(Barrier Method)**

$$\min_{\mathbf{x}} f(\mathbf{x}) - \mu \sum_i \ln(-g_i(\mathbf{x}))$$

$\mu \rightarrow 0$

- **SQP (Sequential Quadratic Programming)** (QP)

- **Lagrange (Augmented Lagrangian)**

(Trust Region Methods)

Δ_k

$$\min_{\|\mathbf{p}\| \leq \Delta_k} m_k(\mathbf{p}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^\top \mathbf{p} + \frac{1}{2} \mathbf{p}^\top \mathbf{B}_k \mathbf{p}$$

ρ_k

Δ_k

$$\rho_k = \frac{f(\mathbf{x}_k) - f(\mathbf{x}_k + \mathbf{p}_k)}{m_k(\mathbf{0}) - m_k(\mathbf{p}_k)}$$



ill-conditioned



Newton

(Convex Sets)

$$C \subseteq \mathbb{R}^n \quad \mathbf{x}, \mathbf{y} \in C \quad \theta \in [0, 1]$$

$$\theta \mathbf{x} + (1 - \theta) \mathbf{y} \in C$$

- $\{\mathbf{x} \mid \mathbf{a}^\top \mathbf{x} = b\}$

- $\{\mathbf{x} \mid \mathbf{a}^\top \mathbf{x} \leq b\}$

- $\{\mathbf{x} \mid \mathbf{A}\mathbf{x} \preceq \mathbf{b}\}$

-

- $\{(\mathbf{x}, t) \mid \|\mathbf{x}\| \leq t\}$

- $\bigcap_{\alpha} C_{\alpha}$

- $f(C)$

- $/$

(Convex Functions)

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\mathbf{x}, \mathbf{y} \quad \theta \in [0, 1]$$

$$f(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta) f(\mathbf{y})$$

- $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$
 - $\nabla^2 f(\mathbf{x}) \succeq \mathbf{0}$ Hessian
 - (Epigraph) $\text{epi}(f) = \{(\mathbf{x}, t) \mid f(\mathbf{x}) \leq t\} \iff f$
- $$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2 \quad \mu > 0$$

$$\begin{aligned}
 \min_{\mathbf{x}} \quad & f_0(\mathbf{x}) \\
 \text{s.t.} \quad & f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\
 & \mathbf{Ax} = \mathbf{b}
 \end{aligned}$$

$f_0, f_1, \dots, f_m$

= KKT

- LP, QP (), QCQP ()
- SOCP (), SDP ()
- (Geometric Programming)

Lagrange

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$$

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_i \lambda_i f_i(\mathbf{x}) + \sum_j \nu_j (\mathbf{a}_j^\top \mathbf{x} - b_j)$$

Slater (Slater's Condition)

Primal = Dual

(Subgradient Method)

f

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \mathbf{g}_k, \quad \mathbf{g}_k \in \partial f(\mathbf{x}_k)$$

$\partial f(\mathbf{x}) =$

- $\alpha_k \rightarrow 0, \sum \alpha_k = \infty$
- $\mathcal{O}(1/\sqrt{k})$
- ℓ_1 SVM

(Proximal Gradient Method)

$$\min_{\mathbf{x}} f(\mathbf{x}) + g(\mathbf{x}) \quad f \text{ diff} \quad g \text{ prox}$$

(Proximal Operator)

$$\text{prox}_{\eta g}(\mathbf{v}) = \arg \min_{\mathbf{x}} \left\{ g(\mathbf{x}) + \frac{1}{2\eta} \|\mathbf{x} - \mathbf{v}\|^2 \right\}$$

ISTA / FISTA /

$$\mathbf{z}_k = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$$

$$\mathbf{x}_{k+1} = \text{prox}_{\eta g}(\mathbf{z}_k)$$

- ℓ_1 (LASSO)
- FISTA $\mathcal{O}(1/k^2)$

Nesterov (Accelerated Gradient)

$\mathcal{O}(1/k)$ Nesterov $\mathcal{O}(1/k^2)$

Nesterov (NAG)

$$\mathbf{y}_k = \mathbf{x}_k + \beta_k(\mathbf{x}_k - \mathbf{x}_{k-1})$$

$$\mathbf{x}_{k+1} = \mathbf{y}_k - \alpha_k \nabla f(\mathbf{y}_k)$$

$$\beta_k = \frac{k-1}{k+2}$$

- "Look ahead"

- $\mathcal{O}((1 - \sqrt{\mu/L})^k)$

- Nesterov SGD with Momentum Adam

ADMM (Alternating Direction Method of Multipliers)

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{z}} \quad & f(\mathbf{x}) + g(\mathbf{z}) \\ \text{s.t.} \quad & \mathbf{Ax} + \mathbf{Bz} = \mathbf{c} \end{aligned}$$

$$\text{Lagrange } \mathcal{L}_\rho(\mathbf{x}, \mathbf{z}, \mathbf{y}) = f(\mathbf{x}) + g(\mathbf{z}) + \mathbf{y}^\top (\mathbf{Ax} + \mathbf{Bz} - \mathbf{c}) + \frac{\rho}{2} \|\mathbf{Ax} + \mathbf{Bz} - \mathbf{c}\|^2$$

ADMM

$$\mathbf{x}_{k+1} = \arg \min_{\mathbf{x}} \mathcal{L}_\rho(\mathbf{x}, \mathbf{z}_k, \mathbf{y}_k)$$

$$\mathbf{z}_{k+1} = \arg \min_{\mathbf{z}} \mathcal{L}_\rho(\mathbf{x}_{k+1}, \mathbf{z}, \mathbf{y}_k)$$

$$\mathbf{y}_{k+1} = \mathbf{y}_k + \rho(\mathbf{Ax}_{k+1} + \mathbf{Bz}_{k+1} - \mathbf{c})$$

(SDP)

SDP LP

$$\begin{aligned} \min_{\mathbf{X}} \quad & \langle \mathbf{C}, \mathbf{X} \rangle = \text{tr}(\mathbf{C}^\top \mathbf{X}) \\ \text{s.t.} \quad & \langle \mathbf{A}_i, \mathbf{X} \rangle = b_i, \quad i = 1, \dots, m \\ & \mathbf{X} \succeq \mathbf{0} \quad (\quad) \end{aligned}$$

- (Primal-Dual IPM)
- PCA SDP
- Max-Cut Goemans–Williamson 0.878 SDP

LP $\subset$ SOCP $\subset$ SDP

- CVXPY (Python)
- CVX (MATLAB)
- Convex.jl (Julia)

- LP $\rightarrow$ LP
- QP $\rightarrow$ QP
- SOCP/SDP $\rightarrow$
- NLP (IPOPT, SNOPT)

- ECOS, SCS —SOCP, SDP
- OSQP —QP
- MOSEK — LP/QP/SOCP/SDP
- Gurobi / CPLEX —LP, MILP, QP

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell(\mathbf{x}_i, y_i; \theta) + \lambda R(\theta)$$

- $\ell(\cdot)$ MSE Hinge
- $R(\cdot)$ ℓ_2 , ℓ_1 , dropout
- N ML

ML

- $N \rightarrow$
- $n \rightarrow$
-
- $\gg$

(SGD)

(mini-batch) $\mathcal{B}$

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \cdot \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla_{\boldsymbol{\theta}} \ell(\mathbf{x}_i, y_i; \boldsymbol{\theta}_t)$$

- $\mathbb{E}[\nabla f_{\mathcal{B}}(\boldsymbol{\theta})] = \nabla f(\boldsymbol{\theta})$
- $\mathcal{O}(1/\sqrt{T}) \quad \mathcal{O}(1/T)$
- batch size

SGD ill-conditioned

SGD with Momentum (Polyak)

$$\begin{aligned}\mathbf{v}_t &= \beta \mathbf{v}_{t-1} + \eta \nabla f(\boldsymbol{\theta}_t) \\ \boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - \mathbf{v}_t\end{aligned}$$

Nesterov Accelerated Gradient (NAG)

$$\begin{aligned}\mathbf{v}_t &= \beta \mathbf{v}_{t-1} + \eta \nabla f(\boldsymbol{\theta}_t - \beta \mathbf{v}_{t-1}) \\ \boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - \mathbf{v}_t\end{aligned}$$



■ NAG "Look-ahead"

■ $\beta = 0.9$

Adam (Adaptive Moment Estimation)

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t$$

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2) \mathbf{g}_t^2$$

$$\hat{\mathbf{m}}_t = \mathbf{m}_t / (1 - \beta_1^t), \quad \hat{\mathbf{v}}_t = \mathbf{v}_t / (1 - \beta_2^t)$$

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \cdot \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon}$$

- $\hat{\mathbf{m}}_t$
- $\hat{\mathbf{v}}_t$ RMS
- $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$
- AdamW weight decay AdaBelief LAMB

(Learning Rate Schedule)

η

- **Step decay** epoch
- **Cosine annealing**

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \frac{t\pi}{T}\right)$$

- **Warmup** epoch η Transformer
- **Cyclic / OneCycle** LR

Warmup

Transformer Warmup warmup Adam

■ **Natural Gradient / K-FAC** Fisher Hessian

■ **Shampoo** Google

■ **AdaHessian** Hutchinson Hessian

■ **Newton Sketch** Newton

Hessian $n \times n$ $\rightarrow$ Hessian

> 100B GPU

(Data Parallel) + All-Reduce

- GPU mini-batch
- All-Reduce

(Model Parallel)

- Tensor Parallelism GPU
- Pipeline Parallelism GPU

- **Gradient Accumulation** batch

- **(FP16/BF16)**

- **ZeRO** (Zero Redundancy Optimizer)

ZeRO (DeepSpeed)

Microsoft DeepSpeed ZeRO GPU

ZeRO-1	(Adam m, v)	4×
ZeRO-2	+	8×
ZeRO-3	+ +	GPU

175B GPT-3 GPU

(Full Fine-tuning)

-
- 16-32 + +

(PEFT)

- **LoRA** $\Delta W = BA$
- **Adapter**
- **Prefix / Prompt Tuning** token

—

RLHF (Reinforcement Learning from Human Feedback)

- 1 SFT (Supervised Fine-Tuning)
- 2 Reward Model Bradley-Terry
- 3 PPO (Proximal Policy Optimization) Reward Model

$$\max_{\theta} \mathbb{E} [r(x, y) - \beta \cdot \text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}})]$$

DPO (Direct Preference Optimization) — RL reward PPO

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

- **(GPTQ, AWQ)** FP16 4-bit 8-bit
 - (OBS)
 - $\min_{\hat{W}} \|WX - \hat{W}X\|_F^2$
- **KV Cache** Key-Value
- **(Speculative Decoding)** token
- **FlashAttention** IO-aware

Pareto

LLM

- **(Loss Spike)**
- FP16/BF16 underflow / overflow
- **(Gradient Clipping)** $\|\mathbf{g}\| > \tau \quad \mathbf{g} \leftarrow \tau \cdot \mathbf{g} / \|\mathbf{g}\|$
- **Z-loss / Logit Regularization** logit
- output embedding LayerNorm

Chinchilla Scaling Law C N token D

$$N_{\text{opt}} \propto C^{0.5}, \quad D_{\text{opt}} \propto C^{0.5}$$

- **Decision-Focused Learning** ML
 - $\rightarrow$ LP/IP
- **Learning to Optimize** RL
 - Pointer Network / Transformer TSP, VRP
 - + ML GNN MILP
- **Convex ML** + Elastic Net
- **Bilevel Optimization**

1

2

KKT /

3

ADMM

4

ML SGD → Momentum → Adam

5

LLM ZeRO PEFT LoRA RLHF

6

ML OR